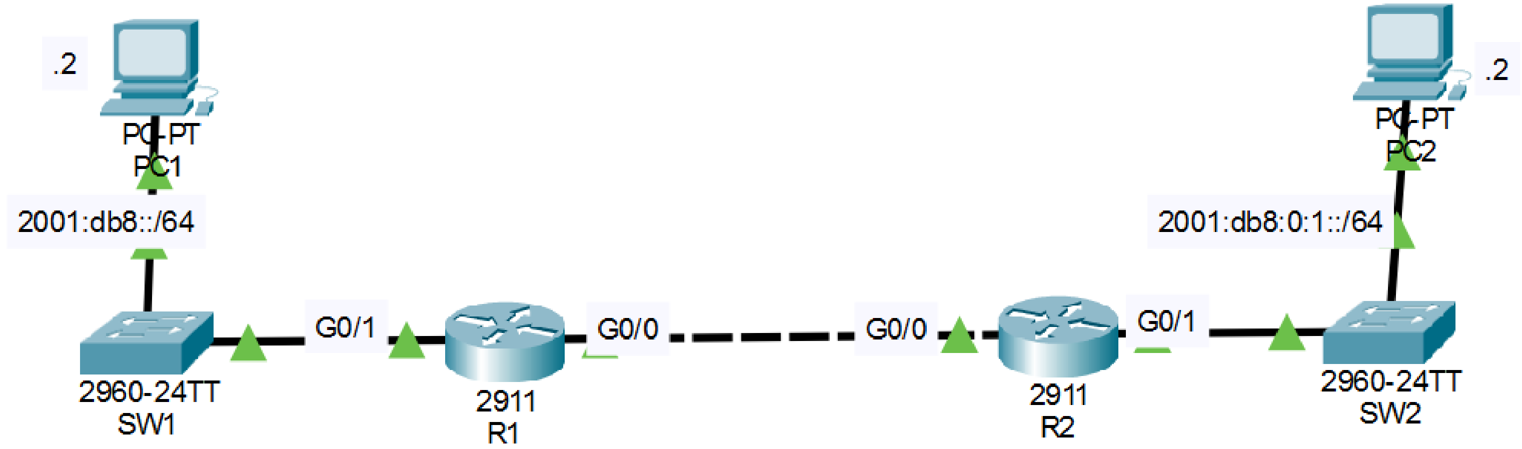
IPv6 EUI-64 and Static Routing setup.



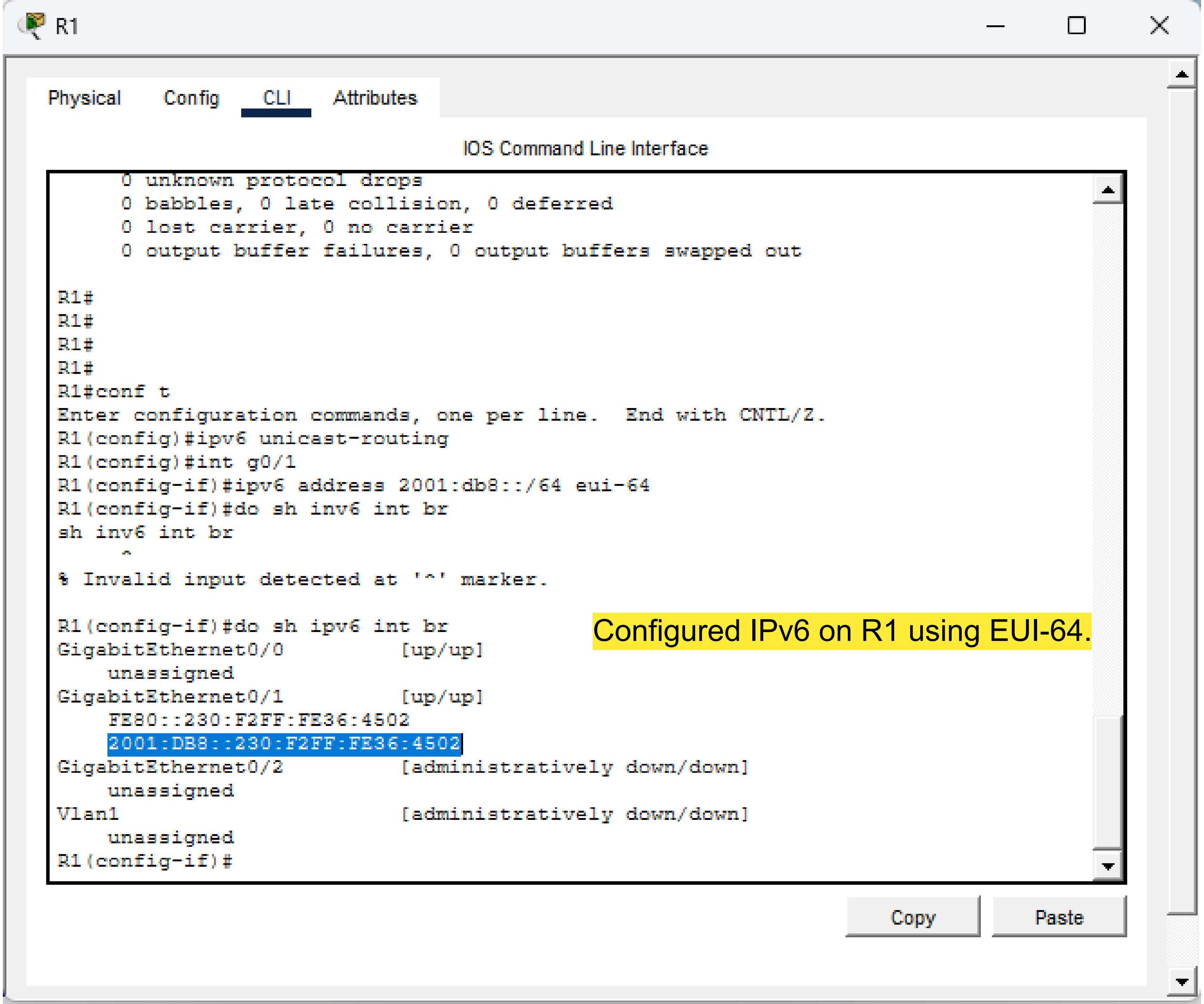
Interfaces are enabled and configured with IPv4.
You will configure IPv6 in the network.

1. Use EUI-64 to configure IPv6 addresses on G0/1 of R1/R2
   *Before configuring the addresses, calculate the EUI-64 interface ID that will be generated on each interface.

2. Configure the appropriate IPv6 addresses/default gateways on PC1 and PC2.

3. Enable IPv6 on G0/0 of R1/R2 without explicitly configuring an IPv6 address.

4. Configure static routes on R1/R2 to enable PC1 to ping PC2.
   Use the 'ipv6 route' command with '?' to learn how to use the command.
   *We will study IPv6 static routes in depth in Day 33.


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#int g0/1
R1(config-if)#ipv6 address 2001:db8::/64 eui-64
R1(config-if)#do sh inv6 int br
sh inv6 int br
^
% Invalid input detected at '^' marker.
R1(config-if)#do sh ipv6 int br
GigabitEthernet0/0 [up/up]
unassigned
GigabitEthernet0/1 [up/up]
FE80::230:F2FF:FE36:4502
2001:DB8::230:F2FF:FE36:4502
GigabitEthernet0/2 [administratively down/down]
unassigned
Vlan1 [administratively down/down]
unassigned
R1(config-if)#
Configured IPv6 on R1 using EUI-64.
Copy
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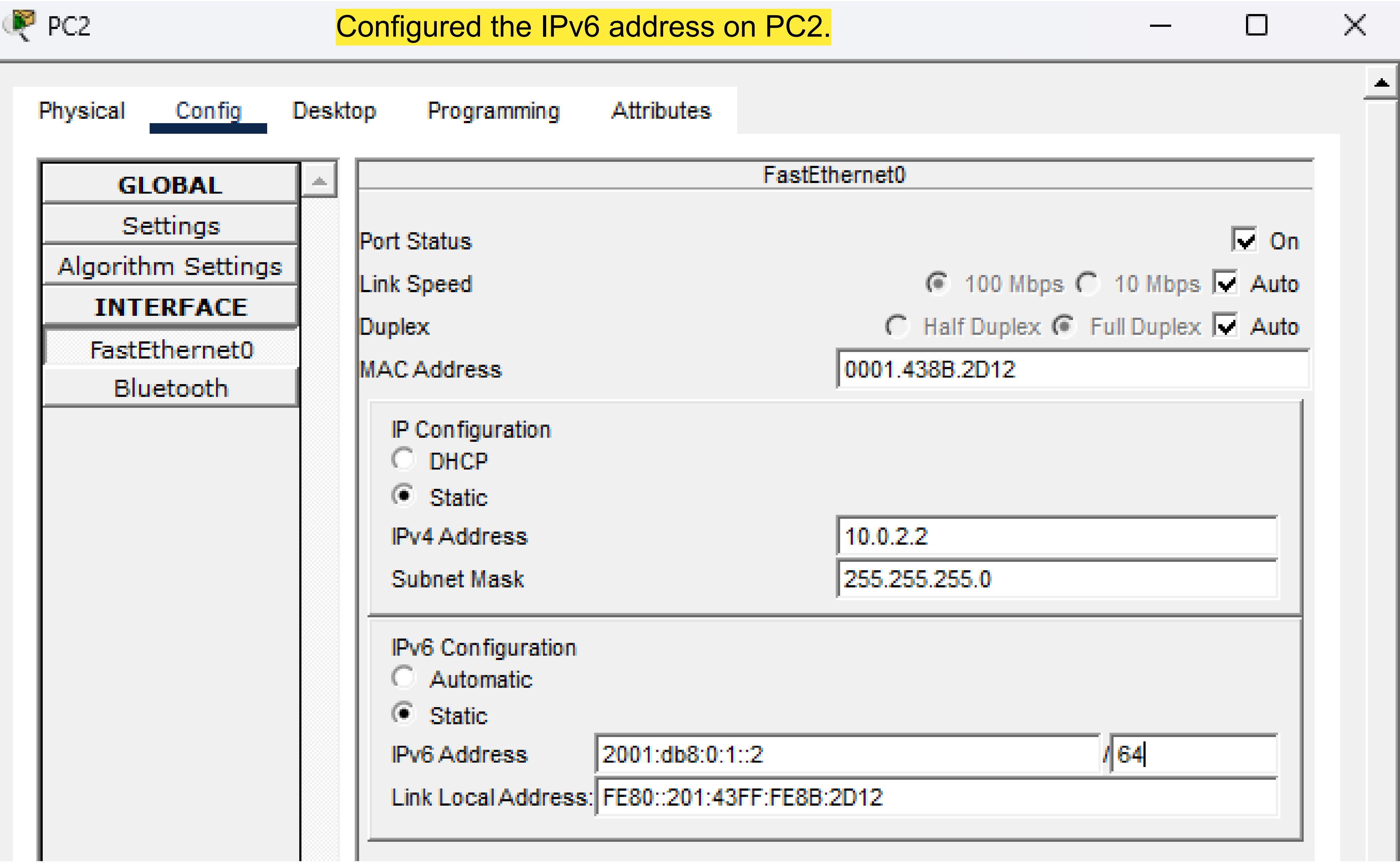
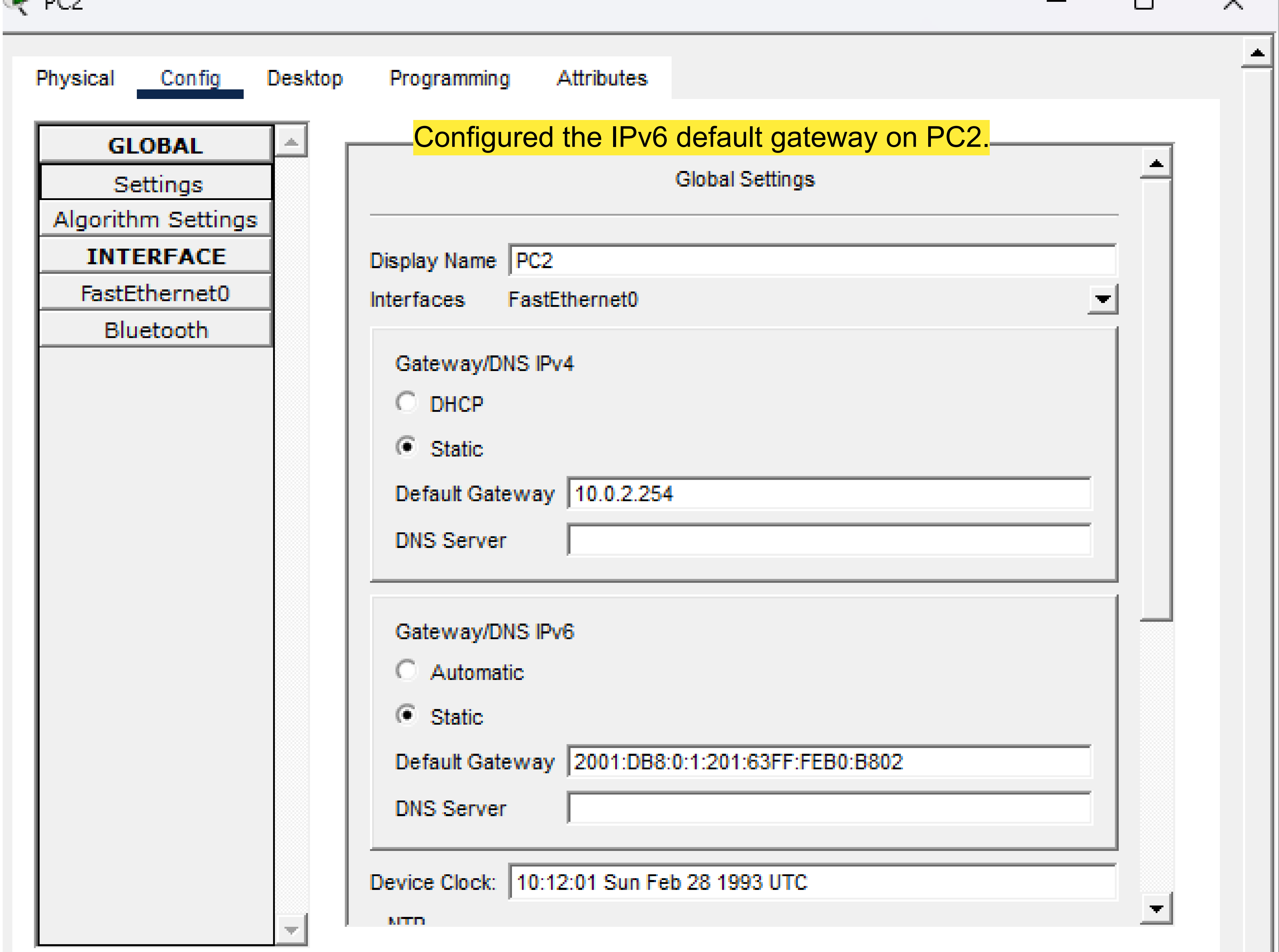

PC2
Configured the IPv6 address on PC2.
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.438B.2D12
IP Configuration
DHCP
Static
IPv4 Address
10.0.2.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:db8:0:1::2
/
64
Link Local Address:
FE80::201:43FF:FE8B:2D12

Configured the IPv6 default gateway on PC2.



R1

Physical Config CLI Attributes

IOS Command Line Interface

Verified the EUI-64 IPv6 address on R1.

```
R1>
R1>
R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ipv6 enable
R1(config-if)#do sh ipv6 int br
GigabitEthernet0/0         [up/up]
    FE80::230:F2FF:FE36:4501
GigabitEthernet0/1         [up/up]
    FE80::230:F2FF:FE36:4502
    2001:DB8::230:F2FF:FE36:4502
GigabitEthernet0/2         [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R1(config-if)#
```

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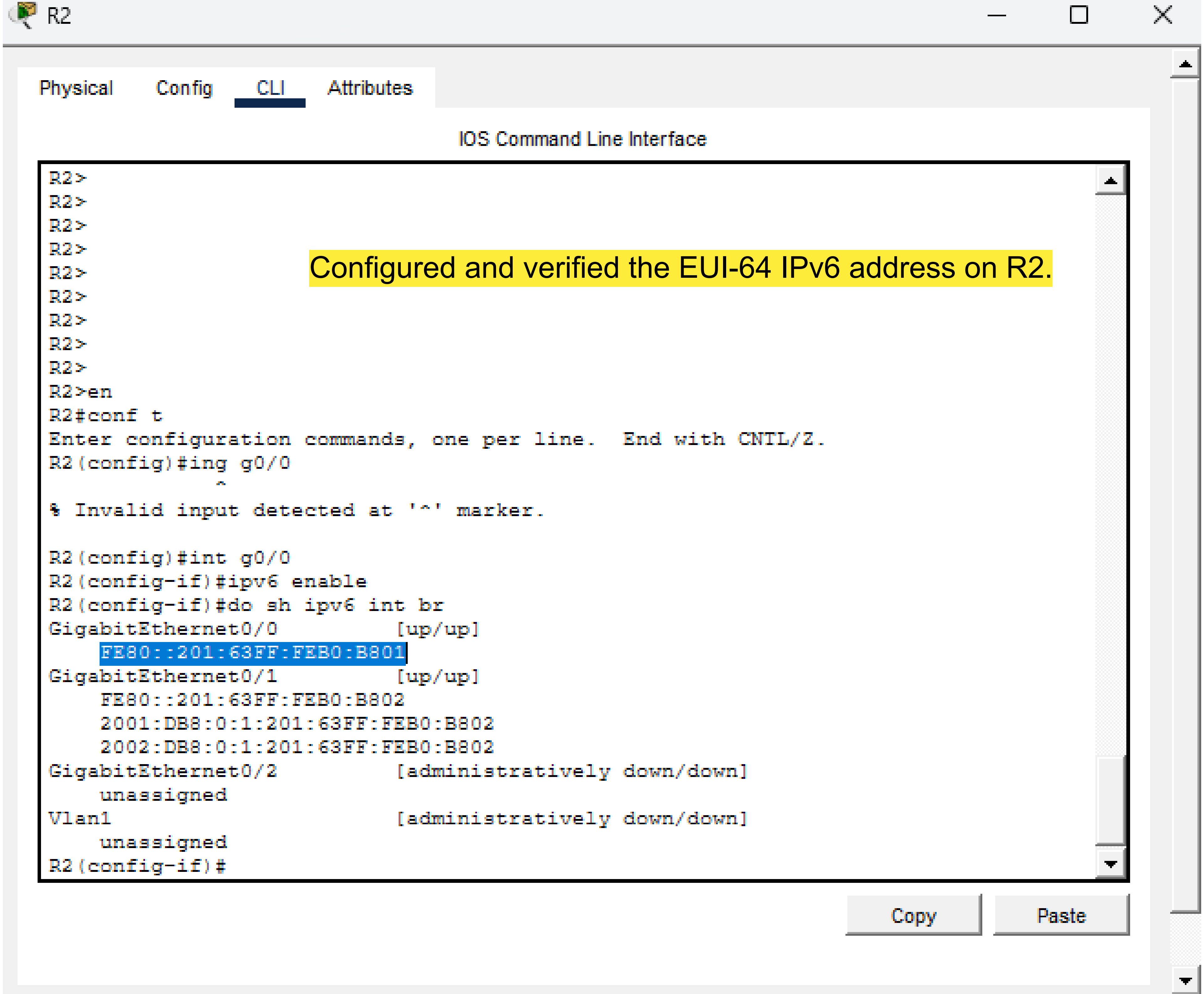
R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Configured and verified the EUI-64 IPv6 address on R2.
R2>
R2>
R2>
R2>
R2>
R2>
R2>
R2>
R2>
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ing g0/0
^
% Invalid input detected at '^' marker.
R2(config)#int g0/0
R2(config-if)#ipv6 enable
R2(config-if)#do sh ipv6 int br
GigabitEthernet0/0 [up/up]
FE80::201:63FF:FEB0:B801
GigabitEthernet0/1 [up/up]
FE80::201:63FF:FEB0:B802
2001:DB8:0:1:201:63FF:FEB0:B802
2002:DB8:0:1:201:63FF:FEB0:B802
GigabitEthernet0/2 [administratively down/down]
unassigned
Vlan1 [administratively down/down]
unassigned
R2(config-if)#
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R1

IOS Command Line Interface

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 route?
route  router
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 route ?
  X:X:X:X::X/<0-128>  IPv6 prefix
R1(config)#ipv6 route ipv6 route 2001:db8:0:1::/64 FE80::201:63FF:FEB0:B801
                      ^
% Invalid input detected at '^' marker.

R1(config)#ipv6 route 2001:db8:0:1::/64 FE80::201:63FF:FEB0:B801
% Interface has to be specified for a link-local nexthop
R1(config)#ipv6 route 2001:db8:0:1::/64 FE80::201:63FF:FEB0:B801 ?
  <1-254>  Administrative distance
  <cr>
R1(config)#ipv6 route 2001:db8:0:1::/64 g0/0 FE80::201:63FF:FEB0:B801
R1(config)#
```

Configured an IPv6 static route on R1.

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Top

R2

IOS Command Line Interface

```
R2>
R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ing g0/0
                ^
% Invalid input detected at '^' marker.

R2(config)#int g0/0
R2(config-if)#ipv6 enable
R2(config-if)#do sh ipv6 int br
GigabitEthernet0/0         [up/up]
    FE80::201:63FF:FEB0:B801
GigabitEthernet0/1         [up/up]
    FE80::201:63FF:FEB0:B802
    2001:DB8:0:1:201:63FF:FEB0:B802
    2002:DB8:0:1:201:63FF:FEB0:B802
GigabitEthernet0/2         [administratively down/down]
    unassigned
Vlan1                      [administratively down/down]
    unassigned
R2(config-if)#
R2(config-if)#
R2(config-if)#
R2(config-if)#
R2(config-if)#
R2(config-if)#exit
R2(config)#ipv6 route 2001:db8::/64 g0/0 FE80::230:F2FF:FE36:4501
R2(config)#
```

Configured an IPv6 static route on R2.

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Top

Verified IPv6 configuration on R1.

```
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#int g0/1
R1(config-if)#ipv6 address 2001:db8::/64 eui-64
R1(config-if)#do sh inv6 int br
sh inv6 int br
     ^
% Invalid input detected at '^' marker.

R1(config-if)#do sh ipv6 int br
GigabitEthernet0/0          [up/up]
    unassigned
GigabitEthernet0/1          [up/up]
    FE80::230:F2FF:FE36:4502
    2001:DB8::230:F2FF:FE36:4502
GigabitEthernet0/2          [administratively down/down]
    unassigned
Vlan1                       [administratively down/down]
    unassigned
R1(config-if)#
```

Successfully tested connectivity between PC1 and PC2 using IPv6.

```
C:\>
C:\>
C:\>ping 2001:db8:0:1::2

Pinging 2001:db8:0:1::2 with 32 bytes of data:

Reply from 2001:DB8:0:1::2: bytes=32 time=11ms TTL=126
Reply from 2001:DB8:0:1::2: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:0:1::2: bytes=32 time<1ms TTL=126
Reply from 2001:DB8:0:1::2: bytes=32 time<1ms TTL=126

Ping statistics for 2001:DB8:0:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```